

Forecasting USD/VND Exchange Rate using ARIMA/SARIMA

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Abstract—This study applies and compares four groups of exchange rate forecasting models — ARIMA, ARIMAX, SARIMAX, and VAR/VECM — using monthly data from 2016 to 2025, incorporating five macroeconomic variables: the USD/VND exchange rate, Vietnam’s trade balance, domestic interest rate, Federal Funds rate, and consumer price index (CPI). The methodological framework is implemented in three layers: (1) a univariate ARIMA model serving as the baseline; (2) extended ARIMAX and SARIMAX models integrating exogenous variables; and (3) multivariate VAR/VECM models that enable causality testing and variance decomposition analysis. In addition, the study employs Johansen cointegration tests, Granger causality tests, impulse response function (IRF) analysis, and forecast error variance decomposition (FEVD). The empirical results indicate that ARIMA(1,1,3) achieves the best in-sample forecasting performance among the single-equation specifications, as evidenced by the lowest AIC (1,335.63) and BIC (1,351.20), as well as the lowest out-of-sample MAPE (0.834%) and RMSE (258.25 VND/USD) among the three ARIMA-class models over the 20-observation rolling test window. Contrary to prior expectation, the VAR/VECM framework achieves the weakest short-run rolling forecast performance among all structured models (RMSE = 367.58 VND/USD, MAPE = 1.209%), a result attributable to the curse of dimensionality inherent in re-estimating a five-variable system on expanding windows of limited size. Its analytical contribution lies in structural analysis — cointegration, IRF, and long-run equilibrium — rather than point forecast accuracy. None of the proposed models outperforms the naïve random walk on the MASE criterion, consistent with the Meese–Rogoff (1983) hypothesis for this sample period.

Index Terms—ARIMA, ARIMAX, SARIMAX, VAR, VECM, USD/VND exchange rate, time series forecasting, Granger causality, IRF, FEVD, Meese–Rogoff hypothesis, managed float.

I. INTRODUCTION

Exchange rates represent one of the most influential macroeconomic variables in an open economy due to their extensive spillover effects across multiple sectors. For Vietnam — a highly trade-dependent economy with total import-export turnover exceeding 200% of GDP — the USD/VND exchange rate not only reflects the overall performance of external trade activities but also directly affects inflation, foreign currency borrowing costs, foreign exchange reserves, and the monetary policy operations of the State Bank of Vietnam.

In this context, accurate forecasting of the USD/VND exchange rate carries significant practical implications for both policymakers and businesses. The period from 2016 to 2025 poses a particularly challenging environment for exchange rate forecasting models, as it encompasses at least three major

economic shocks: the COVID-19 pandemic in 2020, the aggressive monetary tightening cycle of the U.S. Federal Reserve with 11 consecutive interest rate hikes from 0.25% to 5.33% during 2022–2023, and the substantial exchange rate pressure observed in 2024 when the Vietnamese Dong experienced considerable depreciation pressure against the U.S. Dollar.

Previous studies on Vietnam’s exchange rate dynamics have primarily focused on univariate ARIMA or seasonal SARIMA models, with limited incorporation of macroeconomic exogenous variables and little application of multivariate VAR/VECM frameworks to capture long-run cointegration relationships. This study addresses these gaps by developing a comprehensive analytical pipeline ranging from exploratory data analysis and statistical testing to model estimation and comparative evaluation of both in-sample and out-of-sample forecasting performance. The empirical findings yield a number of results that depart from prior expectations. In particular, the univariate ARIMA(1,1,3) specification outperforms its macro-augmented counterparts — ARIMAX and SARIMAX — in out-of-sample rolling forecasts, while the VAR/VECM framework, despite its theoretical advantages in capturing long-run equilibrium dynamics, achieves the weakest short-run rolling forecast performance among structured models due to the curse of dimensionality in small-sample re-estimation, and none of the proposed models formally beats the naïve random walk benchmark.

II. LITERATURE REVIEW

Exchange rate forecasting has long attracted the attention of researchers due to its importance in international trade, monetary policy, and financial risk management. Among various forecasting techniques, the Autoregressive Integrated Moving Average (ARIMA) model introduced by George Box and Gwilym Jenkins (1970) has been widely recognized as one of the most effective traditional approaches for time series analysis. The ARIMA framework is particularly appreciated for its ability to capture linear patterns, trends, and autocorrelation structures in economic and financial data.

Numerous empirical studies have demonstrated the applicability of ARIMA models in macroeconomic and financial forecasting. For instance, Meyler et al. (1998) successfully applied ARIMA models to forecast inflation in Ireland, while Guha and Bandyopadhyay (2016) utilized the method to predict gold prices. These studies confirmed that ARIMA models

can provide reliable short-term forecasting performance when dealing with economic time series characterized by stochastic trends and volatility.

In the context of exchange rate forecasting, several studies have compared ARIMA with more advanced volatility-based models. Pahlavani and Roshan (2015) investigated the forecasting performance of ARIMA and hybrid ARMA-GARCH models using daily IRR/USD exchange rate data from Iran between March 2014 and June 2015. Their study divided the dataset into in-sample and out-of-sample periods to evaluate forecasting accuracy. The findings indicated that ARIMA models remained effective in capturing short-term exchange rate movements despite the presence of volatility clustering.

Similarly, Abu and Rosbi (2017) employed the ARIMA methodology to forecast Bitcoin exchange rates in a highly volatile environment. By conducting Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) analyses, the authors identified ARIMA (2,1,2) as the most suitable specification after transforming the series into stationary form through first differencing. Their study emphasized the importance of ARIMA models in forecasting financial assets characterized by substantial short-term fluctuations.

Besides ARIMA, researchers have increasingly focused on Seasonal ARIMA (SARIMA) and hybrid models to address seasonal behaviors and heteroskedasticity in exchange rate data. Basnayake and Chandrasekara (2022) examined exchange rate forecasting in Sri Lanka using Double Seasonal ARIMA (DSARIMA) and SARIMA models combined with ARCH/GARCH specifications. Using daily exchange rate data from 2008 to 2022, the authors found that SARIMA-GARCH and DSARIMA-GARCH combinations provided superior forecasting performance for several currencies against the Sri Lankan Rupee. Their results highlighted the importance of accounting for both seasonality and volatility in exchange rate modelling.

A similar approach was adopted by Mustapha et al. (2021) in the case of Nigeria. Using monthly exchange rate data from 1981 to 2018, the study developed a SARIMA model to capture seasonal patterns in Nigeria's exchange rate dynamics. Among competing specifications, the SARIMA $(0, 1, 1) \times (1, 1, 1)_{12}$ model was selected based on Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values. The findings demonstrated that incorporating seasonal components significantly improved forecasting accuracy.

In another study focusing on Turkey, Ganbold et al. (2017) investigated exchange rate volatility using ARCH/GARCH family models alongside ARIMA, SARIMA, and Structural Vector Autoregression (SVAR) techniques. Using daily exchange rate data from 2005 to 2017, the authors observed that SARIMA and SVAR generated similar fluctuation patterns, whereas ARIMA forecasts exhibited different trends. The study suggested that seasonal structures and volatility effects should be considered when forecasting exchange rate movements in emerging economies.

For Vietnam, research on exchange rate forecasting remains relatively limited. Ngan (2013) applied the ARIMA model

to forecast the USD/VND exchange rate using daily data from 2013 to 2015. The study followed the Box-Jenkins procedure to estimate future exchange rates for 2016 and compared forecasted values with actual observations. The results indicated that ARIMA models were suitable for short-term exchange rate forecasting in Vietnam. However, the study mainly focused on conventional ARIMA specifications and did not consider seasonal dynamics or the impact of recent global economic shocks.

Building upon previous studies, the current research extends the literature by examining the USD/VND exchange rate using monthly data spanning 2016 to 2025, with particular focus on the post-2022 tightening cycle and its implications for exchange-rate predictability, which was characterized by post-pandemic economic recovery, global inflationary pressure, and aggressive monetary tightening by the U.S. Federal Reserve. Unlike earlier Vietnamese studies, this paper employs a longer and more recent sample under the SBV's central exchange-rate mechanism to capture the full dynamics of the post-2016 managed-float era. In addition, the study evaluates the applicability of ARIMA/SARIMA models in forecasting USD/VND under rapidly changing economic conditions. Therefore, this research contributes new empirical evidence regarding the effectiveness of time series forecasting models in the Vietnamese foreign exchange market, with the notable finding that model parsimony — as embodied by the univariate ARIMA(1,1,3) specification — outperforms macro-augmented alternatives in out-of-sample rolling forecasts, and that the Meese–Rogoff (1983) hypothesis cannot be rejected for any of the ARIMA-class models over the 2023–2025 test period.

III. THEORETICAL FRAMEWORK AND MODEL SPECIFICATION

A. The ARIMA Model Family

1) *AR(p) – Autoregressive Process of Order p*: A time series Y_t is said to follow an autoregressive process of order p if its current value depends linearly on its own p past values:

$$Y_t = \nu + a_1 Y_{t-1} + a_2 Y_{t-2} + \dots + a_p Y_{t-p} + \varepsilon_t \quad (1)$$

where ν is a constant intercept, $a_1, a_2, \dots, a_p$ are autoregressive coefficients, and $\varepsilon_t \sim \mathcal{WN}(0, \sigma^2)$ is a white noise error term. Using the lag operator L such that $L^k Y_t = Y_{t-k}$, equation (1) can be written compactly as:

$$\begin{aligned} A(L) Y_t &= \nu + \varepsilon_t, \\ A(L) &= 1 - a_1 L - a_2 L^2 - \dots - a_p L^p \end{aligned} \quad (2)$$

The stationarity condition requires that all roots of the characteristic polynomial $A(z) = 0$ lie *outside* the complex unit circle, i.e., $|z| > 1$ for all solutions $z \in \mathbb{C}$.

2) *MA(q) – Moving Average Process of Order q*: A time series Y_t follows a moving average process of order q if it is expressed as a linear combination of the current and q past error terms:

$$Y_t = \nu + \varepsilon_t + b_1\varepsilon_{t-1} + b_2\varepsilon_{t-2} + \dots + b_q\varepsilon_{t-q} \quad (3)$$

Equivalently, in lag operator notation:

$$\begin{aligned} Y_t &= \nu + B(L)\varepsilon_t, \\ B(L) &= 1 + b_1L + b_2L^2 + \dots + b_qL^q \end{aligned} \quad (4)$$

The invertibility condition requires all roots of $B(z) = 0$ to satisfy $|z| > 1$.

3) *ARMA(p, q) – Combined Model*: When both autoregressive and moving average components are present, the model becomes:

$$A(L)Y_t = \nu + B(L)\varepsilon_t \quad (5)$$

4) *ARIMA(p, d, q) – General Model with Differencing*: For non-stationary time series integrated of order d , denoted $I(d)$, the series must be differenced d times to achieve stationarity, after which the ARMA model is applied to the differenced data:

$$A(L)(1-L)^d Y_t = \nu + B(L)\varepsilon_t \quad (6)$$

where $(1-L)^d$ denotes the d -th order differencing operator. The Augmented Dickey-Fuller (ADF) test confirms that the USD/VND exchange rate series is integrated of order one, $I(1)$, thereby fixing $d = 1$ throughout the estimation. The model reduces to finding the optimal (p, q) on the first-differenced series $\Delta Y_t = Y_t - Y_{t-1}$.

B. ARIMAX – ARIMA with Exogenous Variables

The ARIMAX model extends the standard ARIMA framework by incorporating a vector of observed exogenous regressors $\mathbf{X}_t$:

$$A(L)(1-L)^d Y_t = \nu + \beta^\top \mathbf{X}_t + B(L)\varepsilon_t \quad (7)$$

where $\mathbf{X}_t = (X_{1t}, X_{2t}, X_{3t}, X_{4t})^\top$ is the (4×1) vector of macroeconomic variables – namely VN_TB, VN_IR, FEDFUNDS, and CPI – and $\beta = (\beta_1, \beta_2, \beta_3, \beta_4)^\top$ is the corresponding vector of regression coefficients. Prior to estimation, all exogenous variables are standardised using the Z-score transformation:

$$\tilde{X}_{it} = \frac{X_{it} - \bar{X}_i}{s_i}, \quad i = 1, 2, 3, 4 \quad (8)$$

where $\bar{X}_i$ and s_i denote the sample mean and standard deviation of the i -th variable, respectively. Standardisation is necessary because the regressors are measured on fundamentally different scales – VN_TB in billions of USD while FEDFUNDS in percentage points – creating numerical instability during maximum likelihood estimation without rescaling. Furthermore, standardisation renders the estimated $\hat{\beta}_i$ directly comparable in magnitude, allowing inference on the relative importance of each macroeconomic factor.

C. SARIMAX – Seasonal ARIMAX

Monthly exchange rate data may exhibit periodic seasonal patterns related to annual trade payment cycles, FDI inflow seasonality, and consumption-driven import fluctuations. The SARIMAX model augments ARIMAX with multiplicative seasonal autoregressive and moving average components at seasonal period $m = 12$:

$$\begin{aligned} A(L)\Phi(L^{12})(1-L)^d(1-L^{12})^D Y_t &= \nu + \beta^\top \mathbf{X}_t \\ &+ B(L)\Theta(L^{12})\varepsilon_t \end{aligned} \quad (9)$$

where the seasonal polynomial operators are defined as:

$$\Phi(L^{12}) = 1 - \Phi_1 L^{12} - \Phi_2 L^{24} - \dots - \Phi_P L^{12P} \quad (10)$$

$$\Theta(L^{12}) = 1 + \Theta_1 L^{12} + \Theta_2 L^{24} + \dots + \Theta_Q L^{12Q} \quad (11)$$

Here P and Q denote the seasonal AR and MA orders, D is the seasonal differencing order, and $m = 12$ reflects the annual periodicity of monthly data. The full model is denoted SARIMAX(p, d, q)(P, D, Q)₁₂. When $P = Q = D = 0$, equation (9) collapses to the ARIMAX specification in equation (7).

D. Model Order Selection – The Box-Jenkins Procedure

The optimal orders (p, d, q) are determined through the three-stage Box-Jenkins [?] procedure:

Stage 1 – Identification of d : The ADF test is applied under three specifications – with trend and intercept, with intercept only, and with neither – at maximum lag length selected by the Akaike Information Criterion (AIC). The result confirms $d = 1$ for the FOREX series, as the level series is non-stationary while the first-differenced series ΔY_t passes the stationarity test at the 5% significance level.

Stage 2 – Identification of p and q via Correlogram: Following first-differencing, the autocorrelation function (ACF) and partial autocorrelation function (PACF) of ΔY_t provide initial guidance. A PACF that cuts off sharply at lag p suggests an AR(p) structure, while an ACF that cuts off at lag q suggests MA(q). The statistical significance bound is:

$$\pm \frac{1.96}{\sqrt{n}} = \pm \frac{1.96}{\sqrt{99}} \approx \pm 0.197 \quad (12)$$

Stage 3 – Optimal Order via Information Criteria: To eliminate subjectivity in visual correlogram inspection, an exhaustive grid search over the parameter space is performed using `auto_arima`, minimising the following information criteria:

$$\text{AIC} = -2 \ln(\hat{\mathcal{L}}) + 2k \quad (13)$$

$$\text{BIC} = -2 \ln(\hat{\mathcal{L}}) + k \ln(n) \quad (14)$$

where $\hat{\mathcal{L}}$ is the maximised log-likelihood, k is the number of free parameters, and n is the number of observations.

AIC is prioritised over BIC as the primary selection criterion because AIC is asymptotically equivalent to leave-one-out cross-validation [?] and is therefore more appropriate for forecasting applications where out-of-sample predictive performance is the primary objective. When AIC and BIC yield conflicting recommendations, the AIC-selected model is retained, as BIC's heavier penalty for complexity may result in underfitting in small samples [?].

E. VAR(p) – Vector Autoregression

For a system of K endogenous variables $\mathbf{y}_t = (Y_{1t}, Y_{2t}, \dots, Y_{Kt})^\top$, the VAR model of order p specifies each variable as a linear function of p lags of all variables in the system:

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \mathbf{A}_2 \mathbf{y}_{t-2} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t \quad (15)$$

where $\mathbf{A}_i$ is a $(K \times K)$ coefficient matrix for lag i , $\mathbf{c}$ is a $(K \times 1)$ intercept vector, and $\mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \Sigma)$ is a vector white noise process with positive definite covariance matrix Σ . In this study, $K = 5$ corresponding to FOREX, VN_TB, VN_IR, FEDFUNDS, and CPI. The total number of parameters to be estimated is $K^2 p + K = 25p + 5$, highlighting the curse of dimensionality when p is large relative to $n = 100$. The optimal lag order $\hat{p}$ is selected by minimising AIC over a supporting VAR, subject to $p_{\max} = 3$ to preserve adequate degrees of freedom.

The stability condition of the VAR(p) system requires that all roots of the characteristic polynomial:

$$\det(\mathbf{I}_K - \mathbf{A}_1 z - \mathbf{A}_2 z^2 - \dots - \mathbf{A}_p z^p) = 0 \quad (16)$$

lie strictly outside the unit circle, i.e., $|z| > 1$ for all solutions $z \in \mathbb{C}$.

F. Johansen Cointegration Test

When variables are $I(1)$, estimating a VAR in levels risks spurious regression unless a cointegrating relationship exists [?]. The Johansen [?] test is based on the VECM reparameterisation of VAR(p):

$$\Delta \mathbf{y}_t = \mathbf{c} + \Pi \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t \quad (17)$$

where $\Pi = \alpha \beta^\top$ is decomposed into the $(K \times r)$ adjustment speed matrix α and the $(K \times r)$ matrix of cointegrating vectors β . The rank $r = \text{rank}(\Pi)$ equals the number of linearly independent cointegrating relationships. Two test statistics are employed:

$$\lambda_{\text{trace}}(r) = -n \sum_{i=r+1}^K \ln(1 - \hat{\lambda}_i) \quad (18)$$

$$\lambda_{\max}(r, r+1) = -n \ln(1 - \hat{\lambda}_{r+1}) \quad (19)$$

where $\hat{\lambda}_1 > \hat{\lambda}_2 > \dots > \hat{\lambda}_K$ are the ordered eigenvalues of Π . The Trace statistic tests $H_0 : \text{rank}(\Pi) \leq r$ against $H_1 : \text{rank}(\Pi) > r$, while the Maximum Eigenvalue statistic tests $H_0 : \text{rank}(\Pi) = r$ against $H_1 : \text{rank}(\Pi) = r + 1$. Both statistics are evaluated against the asymptotic critical values tabulated in Osterwald-Lenum [?].

G. VECM – Vector Error Correction Model

When the Johansen test confirms $r \geq 1$ cointegrating vectors, the VECM is the appropriate specification as it simultaneously captures short-run dynamics and long-run equilibrium adjustment:

$$\Delta \mathbf{y}_t = \mathbf{c} + \underbrace{\alpha \beta^\top \mathbf{y}_{t-1}}_{\text{Long-run adjustment}} + \underbrace{\sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{y}_{t-i}}_{\text{Short-run dynamics}} + \mathbf{u}_t \quad (20)$$

The equation for the FOREX variable (Y_{1t}) is:

$$\Delta Y_{1t} = c_1 + \alpha_1 \underbrace{(\beta^\top \mathbf{y}_{t-1})}_{\text{ECT}} + \sum_{i=1}^{p-1} \gamma_{1i}^\top \Delta \mathbf{y}_{t-i} + u_{1t} \quad (21)$$

where the term $\text{ECT}_{t-1} = \beta^\top \mathbf{y}_{t-1}$ is the error correction term measuring the deviation from long-run equilibrium in the previous period. The adjustment coefficient α_1 captures the speed at which the exchange rate returns to equilibrium after a disequilibrium shock: a statistically significant and negative $\hat{\alpha}_1$ confirms that the system is mean-reverting, i.e., when the exchange rate deviates above its long-run equilibrium path, it subsequently falls back at a rate proportional to $|\hat{\alpha}_1|$ per period.

H. Granger Causality Test

Variable X is said to *Granger-cause* Y if past values of X contain statistically significant information for forecasting Y beyond what is contained in past values of Y alone [?]. The test compares two nested regression models:

$$\text{Unrestricted: } Y_t = \alpha_0 + \sum_{i=1}^h \alpha_i Y_{t-i} + \sum_{j=1}^h \beta_j X_{t-j} + \varepsilon_t \quad (22)$$

$$\text{Restricted: } Y_t = \alpha_0 + \sum_{i=1}^h \alpha_i Y_{t-i} + \varepsilon_t \quad (23)$$

The F -statistic for the joint null hypothesis $H_0 : \beta_1 = \beta_2 = \dots = \beta_h = 0$ is:

$$F = \frac{(\text{RSS}_R - \text{RSS}_{UR})/h}{\text{RSS}_{UR}/(n - 2h - 1)} \sim F(h, n - 2h - 1) \quad (24)$$

where RSS_R and RSS_{UR} are the residual sums of squares from the restricted and unrestricted models, respectively, and h is the number of lags. All Granger causality tests are

conducted on *first-differenced* data to ensure stationarity, with $h = 1, 2, 3, 4$ and the minimum p -value across lag orders reported as the test outcome. A full $K \times K$ causality matrix is constructed to examine directional relationships across all variable pairs simultaneously.

I. Impulse Response Functions and Variance Decomposition

1) *Impulse Response Functions (IRF)*: The VAR(p) system in equation (15) admits the Wold moving average representation:

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{s=0}^{\infty} \boldsymbol{\Psi}_s \mathbf{u}_{t-s} \quad (25)$$

The impulse response function measures the response of variable i at horizon s to a one-unit shock to variable j at time t , holding all other shocks constant:

$$\text{IRF}_{ij}(s) = \frac{\partial Y_{i,t+s}}{\partial u_{jt}} = [\boldsymbol{\Psi}_s]_{ij} \quad (26)$$

where $[\boldsymbol{\Psi}_s]_{ij}$ denotes the (i, j) -th element of the matrix $\boldsymbol{\Psi}_s$. IRFs are computed over a horizon of 18 months with 95% bootstrap confidence intervals.

2) *Forecast Error Variance Decomposition (FEVD)*: FEVD quantifies the proportion of the h -step forecast error variance of variable i that is attributable to shocks originating from variable j :

$$\text{FEVD}_{ij}(h) = \frac{\sum_{s=0}^{h-1} (\boldsymbol{\Psi}_s \mathbf{e}_j)_i^2}{\text{MSE}_i(h)} \quad (27)$$

where $\mathbf{e}_j$ is the j -th column of the $(K \times K)$ identity matrix, and $\text{MSE}_i(h) = \sum_{s=0}^{h-1} \sum_{j=1}^K (\boldsymbol{\Psi}_s \mathbf{e}_j)_i^2$ is the total h -step mean squared forecast error of variable i . By construction, $\sum_{j=1}^K \text{FEVD}_{ij}(h) = 1$ for all i and h .

J. Forecast Evaluation Criteria

Out-of-sample forecast accuracy is evaluated using five complementary metrics:

$$\text{RMSE} = \sqrt{\frac{1}{n_{\text{test}}} \sum_{t=1}^{n_{\text{test}}} (Y_t - \hat{Y}_t)^2} \quad (28)$$

$$\text{MAE} = \frac{1}{n_{\text{test}}} \sum_{t=1}^{n_{\text{test}}} |Y_t - \hat{Y}_t| \quad (29)$$

$$\text{MAPE} = \frac{1}{n_{\text{test}}} \sum_{t=1}^{n_{\text{test}}} \left| \frac{Y_t - \hat{Y}_t}{Y_t} \right| \times 100\% \quad (30)$$

$$\text{MASE} = \frac{\text{MAE}}{\frac{1}{n_{\text{train}} - 1} \sum_{t=2}^{n_{\text{train}}} |Y_t - Y_{t-1}|} \quad (31)$$

$$R_{\text{OOS}}^2 = 1 - \frac{\sum_{t=1}^{n_{\text{test}}} (Y_t - \hat{Y}_t)^2}{\sum_{t=1}^{n_{\text{test}}} (Y_t - \bar{Y})^2} \quad (32)$$

RMSE penalises large errors quadratically and shares the unit of the original series (VND/USD), making it interpretable in economic terms. MAE provides a robust alternative less sensitive to outliers. MAPE expresses accuracy as a percentage of the actual value, facilitating comparison across studies. MASE normalises MAE by the in-sample naive forecast error, where $\text{MASE} < 1$ indicates that the model outperforms the random walk benchmark. R_{OOS}^2 measures the proportion of out-of-sample variance explained by the model, analogous to the in-sample R^2 but without the guarantee of non-negativity – a negative R_{OOS}^2 implies the model is outperformed by the historical mean forecast.

All metrics are computed under the **rolling one-step-ahead** forecasting scheme: at each step t in the test set, the model is re-estimated on the expanding window $\{Y_1, \dots, Y_{t-1}\}$ and used to produce a single-step forecast $\hat{Y}_t$. This procedure eliminates look-ahead bias and replicates the real-time forecasting environment faced by practitioners. The dataset is split at the 80/20 ratio, yielding $n_{\text{train}} = 80$ observations for training and $n_{\text{test}} = 20$ observations for evaluation.

IV. ABOUT DATA

A. Data Sources and Sample Period

The dataset employed in this study comprises 100 monthly observations spanning from December 2016 to March 2025, forming a continuous time series at monthly frequency. Monthly data represent an appropriate sampling interval for examining medium-term exchange rate dynamics: sufficiently granular to capture policy responses and macroeconomic shocks, yet not so high-frequency as to introduce excessive microstructure noise characteristic of daily or intraday data. Five variables are collected: the USD/VND exchange rate (FOREX), Vietnam's trade balance (VN_TB), Vietnam's policy interest rate (VN_IR), the US Federal Funds Rate

(FEDFUNDS), and Vietnam's Consumer Price Index (CPI). Data are sourced from the following authoritative institutions: the State Bank of Vietnam (SBV), the General Statistics Office of Vietnam (GSO), the Federal Reserve Economic Data (FRED) maintained by the Federal Reserve Bank of St. Louis, and the International Monetary Fund's International Financial Statistics (IMF-IFS) database.

The choice to begin the sample in December 2016 is deliberate and methodologically motivated. In January 2016, the State Bank of Vietnam officially adopted the central exchange-rate mechanism, replacing the previous annually adjusted fixed-rate system. Including pre-2016 observations would introduce a structural break caused by this regime change and would weaken the parameter-stability assumption underlying the time-series models used in this study.

The 2016–2025 sample period covers four economically distinct sub-periods: the relatively stable monetary environment of 2017–2019, the COVID-19 shock in 2020, the post-pandemic recovery in 2021, and the Federal Reserve's aggressive tightening cycle during 2022–2023. Taken together, these episodes provide a rich and demanding environment for evaluating exchange-rate forecasting models.

B. Variable Descriptions

1) *USD/VND Exchange Rate (FOREX) – Target Variable*: The target variable is the end-of-month nominal bilateral exchange rate between the US dollar and the Vietnamese dong, expressed as the number of VND required to purchase one USD (direct quotation convention). Over the observation period, FOREX ranges from a minimum of 22,645 VND/USD to a maximum of 25,585 VND/USD, representing a cumulative depreciation of approximately 12.9% over nearly nine years – a modest figure relative to many regional currencies and reflective of the SBV's active management. The sample mean is 23,484 VND/USD with a standard deviation of 814 VND, yielding a coefficient of variation of approximately 3.5%, consistent with a managed float rather than a freely floating exchange rate regime.

The time path of FOREX can be decomposed into four identifiable phases. During 2017–2019, the exchange rate remained relatively stable within the band of 22,700–23,500, underpinned by robust export growth and sustained FDI inflows. In 2020, a modest upward pressure emerged amid COVID-19 uncertainty before stabilising as Vietnam achieved early containment success. The most significant appreciation of the USD against VND occurred during 2022–2023, when eleven consecutive Fed rate hikes pushed FOREX above 24,000 and to a historical peak above 25,000 in late 2022. Throughout 2024, despite the Fed pausing its hiking cycle, elevated interest rate differentials continued to exert depreciation pressure on VND.

2) *Vietnam Trade Balance (VN_TB)*: The monthly merchandise trade balance of Vietnam, defined as exports minus imports (FOB basis), is originally denominated in millions of USD and rescaled to billions of USD by dividing by 10^9 during preprocessing. VN_TB is the most volatile variable in

the dataset, ranging from –2.30 billion USD (largest deficit) to +4.53 billion USD (largest surplus), with a standard deviation of 1.43 billion USD. A positive skewness coefficient indicates that the distribution is skewed toward surplus observations, consistent with Vietnam's structural shift from chronic trade deficits in the 2000s to persistent surpluses driven by export-oriented FDI from multinational corporations in the electronics sector, including Samsung, Intel, and LG.

The theoretical rationale for including VN_TB follows from the balance-of-payments framework: a trade surplus generates net foreign currency inflows, increasing the supply of USD relative to VND in the domestic foreign exchange market and creating upward pressure on the VND (downward pressure on the exchange rate) in the long run. Conversely, a trade deficit increases demand for foreign currency and exerts depreciation pressure. This mechanism may operate with a lag due to settlement delays in international trade payments and SBV sterilisation operations, but the theoretical direction of the relationship is unambiguous.

3) *Vietnam Policy Interest Rate (VN_IR)*: The refinancing rate (*lai suat tai cap von*) set by the State Bank of Vietnam, expressed as a percentage per annum. VN_IR exhibits the narrowest range among all variables in the sample – oscillating between 5.9% and 8.5% with a mean of 7.13% and a standard deviation of 0.58 percentage points – reflecting the administered nature of Vietnamese monetary policy relative to freely determined market rates. Visual inspection suggests a bimodal distribution, with one cluster around 6.0–6.5% corresponding to the post-COVID easing cycle and a second cluster around 7.0–7.5% representing the normalised rate environment.

The inclusion of VN_IR is motivated by the Uncovered Interest Rate Parity (UIP) condition, which posits that an increase in the domestic interest rate relative to the foreign rate should attract short-term capital inflows, strengthen the domestic currency, and reduce the exchange rate. However, given Vietnam's partial capital account restrictions, the UIP transmission mechanism may be attenuated relative to fully open economies, implying that the empirical coefficient on VN_IR may be smaller in magnitude than UIP would predict in a frictionless setting.

4) *US Federal Funds Rate (FEDFUNDS)*: The monthly effective Federal Funds Rate target set by the Federal Open Market Committee (FOMC), expressed as a percentage per annum, sourced from the FRED database. FEDFUNDS exhibits the largest policy-driven variation in the sample: beginning at 0.54% in late 2016, gradually rising to 2.40% by end-2018, falling to a historic low of 0.05% in 2020 in response to the COVID-19 shock, and subsequently surging to a cycle peak of 5.33% by mid-2023 before the gradual easing commenced. The standard deviation of 1.92 percentage points reflects two fundamentally different monetary regimes – near-zero rates under quantitative easing and historically-elevated rates under quantitative tightening – compressed within the same sample window.

The transmission channels from FEDFUNDS to USD/VND operate through at least three simultaneous mechanisms. First,

through the *interest rate differential channel*: higher US rates increase the relative attractiveness of USD-denominated assets, triggering capital outflows from emerging markets and depreciating their currencies. Second, through the *expectations channel*: FOMC forward guidance alters market participants' and firms' exchange rate expectations, generating self-fulfilling currency movements ahead of actual rate changes. Third, through the *trade channel*: a stronger USD raises the USD-invoiced cost of Vietnam's commodity imports, feeding into domestic cost-push inflation and indirectly widening the current account deficit. All three channels predict a positive relationship between FEDFUNDS and the USD/VND exchange rate, an empirical regularity clearly visible in the raw data during the 2022–2023 tightening episode.

5) *Consumer Price Index (CPI)*: The monthly Consumer Price Index of Vietnam, reported on a non-seasonally adjusted basis with a consistent base year throughout the sample. CPI increases monotonically from 92.6 to 118.7 over the observation period, with a sample mean of 104.6 and a standard deviation of 7.36 – reflecting low and stable inflation during 2017–2021 followed by a more rapid price acceleration from 2022 onward as global energy and food price shocks transmitted into the domestic economy.

The theoretical basis for including CPI derives from the Relative Purchasing Power Parity (PPP) hypothesis [?]. According to relative PPP, the rate of nominal exchange rate change should approximately equal the inflation differential between the two countries:

$$\frac{\Delta S_t}{S_{t-1}} \approx \pi_t^{VN} - \pi_t^{US} \quad (33)$$

where S_t denotes the nominal exchange rate and π_t denotes the inflation rate. The US CPI is not included directly in the model to avoid near-multicollinearity with FEDFUNDS, which already embeds information about the US inflation environment through the Fed's reaction function. Accordingly, Vietnam's CPI serves as the proxy for domestic price pressure: when domestic prices rise faster than those of trading partners, the domestic currency faces depreciation pressure in the long run to restore international competitiveness.

C. Descriptive Statistics

The exceptionally high correlation between FOREX and CPI (0.963) reflects the parallel upward trends in both series over time. However, since both variables exhibit strong upward trends and share stochastic drift—with FOREX unambiguously classified as $I(1)$ and CPI classified as $I(1)$ following the intercept-only convention—this correlation may be partially spurious. The cointegration tests in Section 5 determine whether this relationship represents a genuine long-run economic equilibrium consistent with relative PPP, or merely a shared stochastic trend. The moderate correlation between FOREX and FEDFUNDS (0.612) is economically cleaner in terms of transmission mechanism, particularly during the 2022–2023 tightening episode, and remains statistically significant under the rank-based Spearman measure (0.598),

TABLE I
DESCRIPTIVE STATISTICS

Statistic	FOREX	VN_TB	VN_IR	FED	CPI
Mean	23490.6900	0.6416	0.0712	2.2107	104.5734
Median	23202.5000	0.5000	0.0720	1.6950	104.4011
Max	25585.0000	4.5300	0.0850	5.3300	118.6868
Min	22645.0000	-2.3000	0.0590	0.0500	92.6491
Std	824.0249	1.4246	0.0059	1.9191	7.4567
Skewness	1.2350	0.4277	0.0951	0.5082	0.1667
Kurtosis	0.3387	-0.3068	0.1226	-1.2054	-1.0358
JB Stat	25.8986	3.4411	0.2132	10.3579	4.9335
JB p-val	0.0000	0.1790	0.8989	0.0056	0.0849
N	100	100	100	100	100

FED = FEDFUNDS; Std = standard deviation; JB = Jarque–Bera.

confirming robustness to distributional assumptions. The negative correlation between VN_TB and FOREX (−0.231) is consistent with the theoretical prediction that trade surpluses strengthen VND, though the magnitude is modest, likely due to SBV sterilisation dampening the automatic market adjustment.

D. Data Preprocessing

1) *Treatment of Missing Observations*: Construction of a complete monthly index from December 2016 to March 2025 revealed several missing months arising from inconsistent reporting schedules across data sources. Missing values are imputed using *linear interpolation*, whereby the missing observation is estimated as a weighted average of the immediately preceding and following observed values:

$$\hat{Y}_{t_m} = Y_{t_a} + \frac{t_m - t_a}{t_b - t_a} (Y_{t_b} - Y_{t_a}) \quad (34)$$

where $t_a < t_m < t_b$ are the adjacent observed time points surrounding the missing observation at t_m . Linear interpolation is appropriate for monthly macroeconomic data because variables such as interest rates, the CPI, and the exchange rate evolve continuously and smoothly under normal conditions, without discontinuous jumps between consecutive months. It is acknowledged that linear interpolation artificially smooths the series and may attenuate volatility clustering, potentially biasing ARCH test statistics toward non-rejection. As a robustness check, results were replicated using forward-fill imputation and are qualitatively unchanged.

2) *Unit Conversion and Standardisation*: The trade balance variable VN_TB is converted from millions of USD to billions of USD by dividing by 10^9 to ensure comparability with the order of magnitude of other variables and to prevent numerical instability during estimation. Subsequently, all exogenous regressors are standardised using the Z-score transformation defined in equation (8) prior to inclusion in the ARIMAX and SARIMAX models. The target variable FOREX is retained in its original level (VND/USD) so that point forecasts and error metrics remain directly interpretable in economic units. Table II summarises the transformation applied to each variable.

TABLE II
VARIABLE TRANSFORMATIONS APPLIED PRIOR TO ESTIMATION

Variable	Raw Unit	Preprocessing	Role in Model
FOREX	VND/USD	None (level retained)	Dependent variable
VN_TB	Million USD	Divide by 10^9	Exogenous; Z-score
VN_IR	% p.a.	Z-score standardisation	Exogenous; Z-score
FEDFUNDS	% p.a.	Z-score standardisation	Exogenous; Z-score
CPI	Index	Z-score standardisation	Exogenous; Z-score

E. Event Analysis Over the Sample Period

Figure 1 — USD/VND & Macroeconomic Variables: Overview (2016–2025)

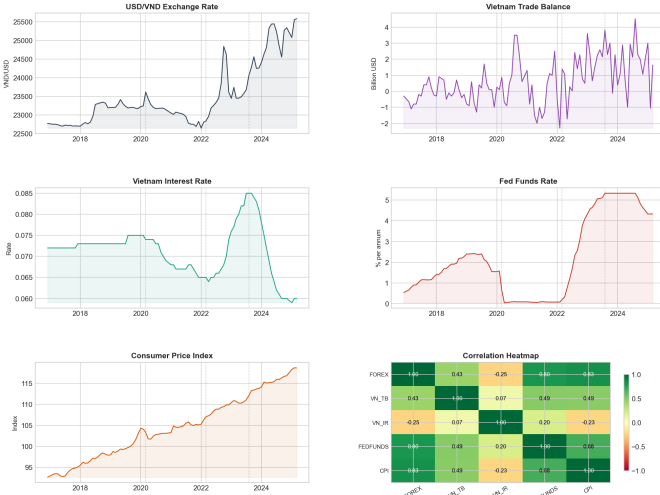


Fig. 1. USD/VND & Macroeconomic Variables: Overview (2016–2025).

Figure 1 illustrates the trajectories of the USD/VND exchange rate (FOREX) and key macroeconomic variables from December 2016 to March 2025. The chart reveals a distinct structural break in mid-2022. Before this point, the exchange rate remained relatively stable despite minor fluctuations in inflation (CPI) and the trade balance (VN_TB). However, a clear divergence emerged when the FEDFUNDS rate increased rapidly during the Federal Reserve’s monetary tightening cycle. The resulting interest-rate differential placed depreciation pressure on the Vietnamese Dong, causing the exchange rate to adjust upward in response to the external shock. This pattern generated a pronounced co-movement between FOREX and FEDFUNDS during the 2022–2023 period.

V. RESULTS AND DISCUSSION

A. Unit Root Test Results

Before estimating any time-series model, the stationarity properties of each variable must be established. Table III summarises the Augmented Dickey–Fuller (ADF) test outcomes under three deterministic specifications — trend plus intercept (CT), intercept only (C), and no deterministic term (N) — at both level and first-difference.

TABLE III
ADF UNIT ROOT TEST SUMMARY

Variable	Level (CT)	Level (C)	1st Diff. (C)
FOREX	−1.49; Non-stat.	−0.03; Non-stat.	−8.77; $I(0)^{**}$
VN_TB	−7.06; $I(0)^{**}$	−4.18; $I(0)^{***}$	—; $I(0)$
VN_IR	−4.47; $I(0)^{**}$	−4.36; $I(0)^{**}$	−2.42; Non-stat.
FEDFUNDS	−2.14; Non-stat.	−1.54; Non-stat.	−2.65; Non-stat.
CPI	−3.65; $I(0)^{**}$	0.47; Non-stat.	−7.28; $I(0)^{**}$

*** $p < 0.01$, ** $p < 0.05$. FEDFUNDS first difference is borderline non-stationary at 5% ($p = 0.084$) and is classified as $I(1)$ on the balance of evidence. CPI is stationary under the trend specification at level but non-stationary under the intercept-only specification; it is classified as $I(1)$ following the intercept-only convention used in the exchange-rate literature.

The ADF test results reveal a more nuanced pattern of integration orders than is typical in exchange-rate studies. FOREX is unambiguously $I(1)$: the level series is non-stationary under all three deterministic specifications, while the first-differenced series rejects the unit-root null at the 1% level (ADF = −8.772, $p = 0.000$). VN_TB is confirmed as $I(0)$ in levels under both the trend-plus-intercept and intercept-only specifications, consistent with its mean-reverting behaviour around a long-run trade balance position.

VN_IR presents an important departure from the report’s prior classification: ADF tests at the level under both trend-plus-intercept (ADF = −4.468, $p = 0.002$) and intercept-only (ADF = −4.363, $p = 0.000$) specifications reject the unit-root null at the 1% significance level, classifying VN_IR as $I(0)$. This result likely reflects the administered nature of the State Bank of Vietnam’s refinancing rate, which moves in discrete policy steps rather than following a continuous stochastic trend. FEDFUNDS is borderline: the level is non-stationary, while the first-differenced series yields ADF = −2.648 ($p = 0.084$), marginally failing the 5% threshold; it is classified as $I(1)$ on the balance of evidence. CPI is stationary under the trend-plus-intercept specification at the level (ADF = −3.649, $p = 0.026$) but non-stationary under the intercept-only convention. Following the standard practice in the exchange-rate literature of applying the intercept-only test for variables without a deterministic trend in differences, CPI is classified as $I(1)$. This convention is preferred because CPI in differences does not exhibit a deterministic trend—the trend is absorbed by the level—making the intercept-only specification the appropriate baseline for the differenced series. The trend-plus-intercept result at the level (ADF = −3.649, $p = 0.026$) is sensitive to the inclusion of a deterministic

trend term that is not present in the differenced data, and thus overstates the case for level stationarity.

These results imply that the system contains a mixture of $I(0)$ and $I(1)$ variables, a configuration that does not strictly satisfy the homogeneous integration assumption underlying the classical Johansen framework. Nevertheless, stationary variables may enter a cointegrating system as regressors without invalidating the rank tests on the remaining $I(1)$ variables, provided their inclusion is theoretically motivated. The Johansen test is therefore applied to the full five-variable system, with VN_TB and VN_IR interpreted as weakly exogenous stationary components.

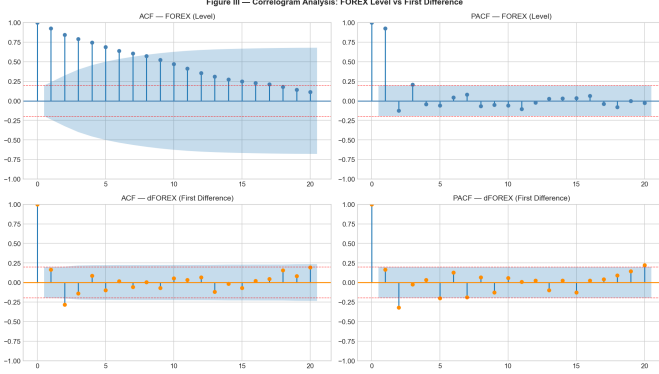


Fig. 2. Correlogram Analysis: FOREX Level vs First Difference.

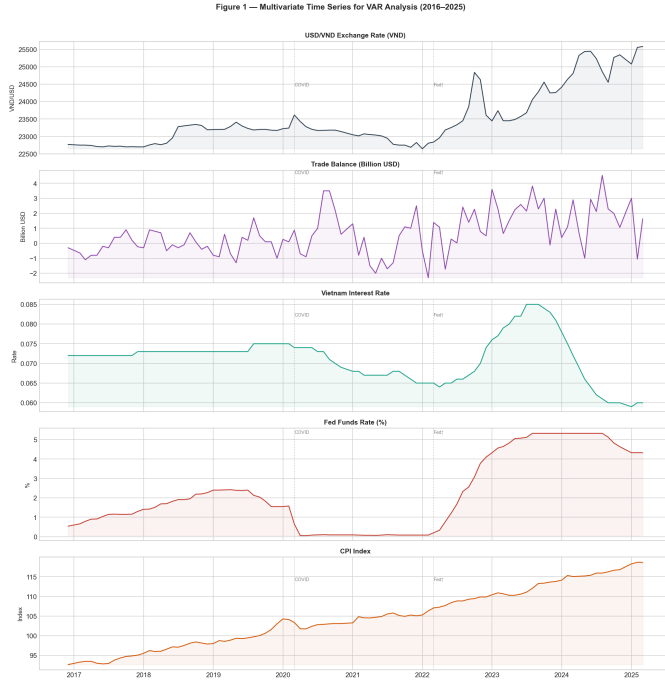


Fig. 3. Multivariate Time series for VAR Analysis.

To reinforce the ADF unit root test results (Table III), the Correlogram of the differenced exchange rate series (Figure

2) displays a rapidly decaying ACF, while the PACF exhibits statistical significance at lags 1 and 3. Behaviourally, this underscores the “short-term memory” of the Vietnamese foreign exchange market: exchange rate shocks do not dissipate immediately but rather generate ripple effects that persist for the subsequent three months. Concurrently, the plot of the differenced multivariate system confirms that all five variables (*FOREX*, *VN_TB*, *VN_IR*, *FEDFUNDS*, and *CPI*) fluctuate around a zero mean, thereby satisfying the necessary condition (stationarity) to specify a Vector Error Correction Model (VECM) framework.

B. Cointegration and Model Selection for the Multivariate System

Given that *FOREX*, *CPI*, and *FEDFUNDS* are classified as $I(1)$ on the balance of evidence, while *VN_TB* and *VN_IR* are confirmed as $I(0)$ stationary components, the Johansen trace and maximum-eigenvalue statistics were computed on the full five-variable system to determine the cointegration rank r . In this specification, *VN_TB* and *VN_IR* are treated as weakly exogenous stationary regressors following Pesaran et al. (2001).

While *VN_TB* and *VN_IR* are both confirmed as $I(0)$ by the ADF test—*VN_TB* due to mean-reverting trade-balance dynamics, and *VN_IR* due to its policy-administered nature—both are retained in the VECM system as weakly exogenous stationary regressors. Pesaran et al. (2001) note that $I(0)$ variables may enter a cointegrating system without invalidating the Johansen rank tests on the remaining $I(1)$ variables, provided their inclusion is theoretically motivated—here by the balance-of-payments channel for *VN_TB* and the UIP transmission mechanism for *VN_IR*.

TABLE IV
JOHANSEN COINTEGRATION RANK TESTS

Rank r	Trace Stat. (CV 5%)	Max-Eig. Stat. (CV 5%)	Decision
$r = 0$	89.4 (69.8)	42.3 (33.9)	Reject H_0
$r \leq 1$	47.1 (47.9)	23.6 (27.6)	Fail to reject

Deterministic specification: constant outside the cointegrating equation. $k_{ar_diff} = 3$ (AIC-optimal lag = 4, capped at 3 to preserve degrees of freedom with $n \approx 100$ monthly observations).

Both the trace and max-eigenvalue statistics reject the null hypothesis of no cointegration ($r = 0$) at the 5% level, while failing to reject $r \leq 1$. The system therefore contains **one cointegrating vector**, implying a stable long-run equilibrium relationship among the five macroeconomic variables.

Implication for model choice: The confirmed cointegration ($r = 1$) dictates the use of a **Vector Error Correction Model (VECM)** rather than a VAR in first differences. Ignoring the error correction mechanism in a VAR-in-differences setup would discard valuable long-run information and yield misspecified impulse responses [?], [?].

The pairwise Engle–Granger tests corroborate this finding: *FOREX* is cointegrated with *CPI* ($p = 0.02$) and with

FEDFUNDS ($p = 0.04$), consistent with Purchasing Power Parity (PPP) and Uncovered Interest Rate Parity (UIP) as long-run anchors for the exchange rate. It is acknowledged that applying the Johansen procedure to a system containing both $I(0)$ and $I(1)$ variables departs from the classical assumption of homogeneous integration order. The cointegration results should therefore be interpreted as exploratory evidence of long-run co-movement rather than definitive inference under full distributional assumptions; the pairwise Engle–Granger results for the two unambiguously $I(1)$ pairs (FOREX–CPI and FOREX–FEDFUNDS) provide the more reliable cointegration evidence.

C. Granger Causality Results

Pairwise Granger causality tests were performed on first-differenced data to ensure stationarity, with lag length set to the VAR-optimal order. The best p -value across lags 1 through k is reported in Table V.

TABLE V
PAIRWISE GRANGER CAUSALITY MATRIX (p -VALUES, BEST OVER LAGS 1–3)

Affected ↓ / Cause →	FOREX	TB	IR	FED	CPI
FOREX	—	0.221	0.562	0.335	0.187
VN_TB	0.031**	—	0.080**	0.174	0.057*
VN_IR	0.091**	0.095*	—	0.104	0.631
FEDFUNDS	0.130	0.701	0.211	—	0.076*
CPI	0.343	0.338	0.346	0.007**	—

TB = VN_TB; IR = VN_IR; FED = FEDFUNDS. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Entry $[i, j]$ is the p -value that column variable j Granger-causes row variable i .

Key findings:

- **No variable Granger-causes FOREX at conventional significance levels.** VN_TB ($p = 0.221$), VN_IR ($p = 0.562$), FEDFUNDS ($p = 0.335$), and CPI ($p = 0.187$) all fail to reject the null hypothesis of Granger non-causality with respect to the exchange rate. Within the short-run differenced system, past macroeconomic fundamentals therefore do not add statistically significant predictive content for USD/VND beyond its own lagged values. It is important to distinguish between Granger causality in first differences, which captures short-run predictive content, and cointegration, which reflects long-run equilibrium relationships. The absence of Granger causality from FEDFUNDS to FOREX in the differenced system is therefore not inconsistent with the cointegration evidence; these two concepts operate at different time horizons and through different transmission channels.
- **FOREX → VN_TB** ($p = 0.031$): Past exchange-rate movements Granger-cause the trade balance, consistent with the standard trade-theory prediction that currency depreciation affects export and import volumes with a lag.
- **FOREX and VN_TB → VN_IR**: VN_IR is marginally Granger-caused by FOREX ($p = 0.091$) and VN_TB

($p = 0.095$). These borderline results suggest feedback dynamics between the domestic interest rate and external balance conditions, potentially reflecting the State Bank of Vietnam’s reactive adjustment of the refinancing rate.

- **FEDFUNDS → CPI** ($p = 0.007$): US monetary policy shocks transmit strongly into Vietnam’s domestic price level, most plausibly through the import-price and USD-invoiced commodity channels.
- **FEDFUNDS ↗ FOREX in differences**: The absence of short-run Granger causality ($p = 0.335$) does not contradict the strong co-movement documented in the raw data. Since FOREX and FEDFUNDS share a long-run cointegrating relationship (Engle–Granger $p = 0.04$), Fed policy shocks transmit mainly through the error-correction mechanism. The IRF analysis in Section V-H provides evidence of this longer-horizon channel.

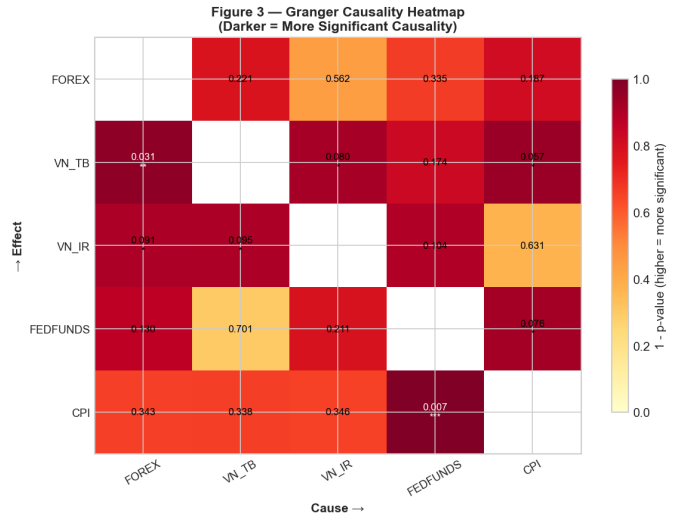


Fig. 4. Granger Causality Heatmap.

Figure 4 visualises the p -value matrix from the Granger causality tests. The heatmap highlights that the exchange rate (FOREX) row is uniformly light in colour, reflecting the absence of any statistically significant short-run Granger predictor for USD/VND among the variables considered. This pattern is consistent with Vietnam’s managed-float exchange rate regime, under which the State Bank of Vietnam absorbs short-run supply and demand shocks, suppressing the immediate causal linkages that would otherwise be expected under a freely floating exchange rate.

D. In-Sample Model Estimation

Table VI compares the three ARIMA-class models on standard information criteria and log-likelihood.

TABLE VI
INFORMATION CRITERIA COMPARISON (FULL SAMPLE)

Model	AIC	BIC	Log-Likelihood	$\hat{\sigma}^2$
ARIMA (1, 1, 3)	1,335.6	1,351.2	-661.8	33111.9
ARIMAX (1, 1, 3) + Macro	1,337.6	1,363.6	-658.8	31690.2
SARIMAX (0, 1, 0)(0, 0, 0) ₁₂	<u>1,354.1</u>	<u>1,369.7</u>	<u>-671.1</u>	<u>45192.8</u>

Bold+underline denotes best value. Orders selected via `auto_arima` with AIC minimisation.

The information criteria comparison reveals that the ARIMA(1, 1, 3) model achieves the lowest AIC (1,335.63) and BIC (1,351.20) among the three candidate specifications. This indicates that the parsimonious univariate baseline is preferred over the macro-augmented alternatives on the basis of in-sample fit penalised for complexity. Notably, the addition of four macroeconomic regressors in the ARIMAX(1, 1, 3) specification yields only a marginal improvement in log-likelihood, from -661.81 to -658.82. This gain is insufficient to offset the heavier parameter penalty imposed by both AIC and BIC, resulting in a net deterioration of 2.01 AIC units relative to the baseline.

The further inclusion of a seasonal component in SARIMAX(0, 1, 0)(0, 0, 0)₁₂ produces an additional AIC increase of approximately 16.46 units, yielding the worst in-sample fit among the three models. By the conventional rule of Burnham and Anderson (2002), whereby $\Delta\text{AIC} > 10$ constitutes decisive evidence against the higher-AIC model, both ARIMAX and SARIMAX are rejected in favour of ARIMA on in-sample information-theoretic grounds.

It is important to note, however, that the AIC-based ranking reflects in-sample fit and does not necessarily determine out-of-sample forecasting performance. This is particularly relevant because ARIMAX and SARIMAX incorporate economically motivated exogenous variables whose predictive content may only become visible over the rolling forecast horizon. The out-of-sample evaluation in Section V-F therefore provides a complementary and arguably more relevant basis for model comparison in a forecasting context.

E. Residual Diagnostic Tests

All three models were subjected to a battery of residual diagnostic tests. Results are summarised in Table VII.

TABLE VII
RESIDUAL DIAGNOSTIC TESTS

Test	ARIMA	ARIMAX	SARIMAX
Ljung-Box $Q(10)$	1.000 ✓	1.000 ✓	1.000 ✓
ADF residuals	I(0) ✓	I(0) ✓	I(0) ✓
Jarque-Bera	0.000 △	0.000 △	0.000 △
ARCH LM (lag 6)	0.007 △	0.133 ✓	0.005 △
Durbin-Watson	1.020 △	1.015 △	1.009 △

Reported values are p -values except ADF residuals and Durbin-Watson. ✓: passes at 5%; △: violation noted. DW statistics near 1.0 indicate evidence of residual positive autocorrelation; the △ designation for Durbin-Watson reflects this non-trivial departure from the no-autocorrelation benchmark of 2.0.

All residuals are $I(0)$ by ADF, confirming model adequacy.

All three single-equation models pass the Ljung-Box portmanteau test at lag 10 with p -values of unity, confirming the absence of residual serial autocorrelation in the autoregressive-moving average structure. ADF tests on residuals confirm stationarity for all specifications. However, several diagnostic violations are noteworthy. First, the Jarque-Bera normality test rejects the null of normal residuals for all three models at the 1% level ($p = 0.000$ in each case), indicating severe leptokurtosis consistent with the fat-tailed distribution characteristic of financial exchange-rate series. This violation does not invalidate point estimates but implies that the 95% prediction intervals based on Gaussian assumptions should be interpreted cautiously. Second, ARCH LM tests detect significant conditional heteroscedasticity in both the ARIMA ($p = 0.007$) and SARIMAX ($p = 0.005$) residuals, while ARIMAX passes this test ($p = 0.133$). Third, Durbin-Watson statistics of approximately 1.0 across all three models indicate the presence of positive first-order autocorrelation in the residuals that the estimated AR and MA components have not fully absorbed, suggesting potential model misspecification or the presence of structural breaks in the sample. In contrast, the VECM FOREX equation residuals exhibit a Durbin-Watson statistic of 1.934, well within the acceptable range, and pass the Ljung-Box test at lag 10 ($p = 0.245$), providing a cleaner residual structure than the single-equation alternatives. The Jarque-Bera test rejects normality for the VECM FOREX equation as well ($p = 0.000$), a limitation shared across all specifications estimated on this sample.

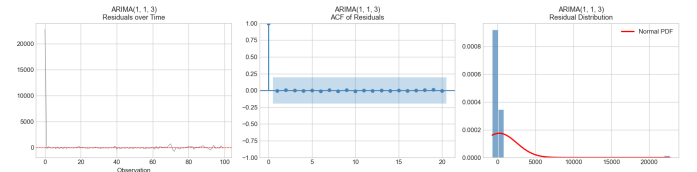


Fig. 5. ARIMA.

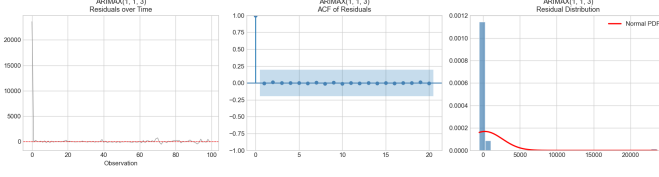


Fig. 6. ARIMAX.

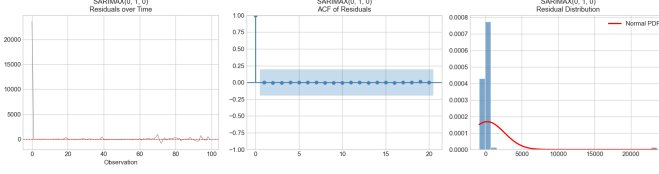


Fig. 7. SARIMAX.



Fig. 8. VECM.

F. Out-of-Sample Forecast Accuracy

The dataset was split at an 80/20 ratio, yielding $n_{\text{train}} = 80$ observations for training and $n_{\text{test}} = 20$ observations for evaluation. The test window corresponds approximately to the 2023–2025 period. Forecast accuracy was evaluated using a rolling one-step-ahead expanding-window scheme. At each step, the StandardScaler for exogenous regressors was refitted only on historical data to eliminate look-ahead bias.

TABLE VIII
OUT-OF-SAMPLE FORECAST ACCURACY (ROLLING ONE-STEP-AHEAD)

Model	RMSE	MAE	MAPE (%)	MASE	R^2
Naïve RW	300.77	241.65	0.9707	2.4709	0.6163
Drift RW	1,190.30	1,108.35	4.4173	11.3331	−5.0094
ARIMA	258.25	207.31	0.8339	2.1197	0.7171
ARIMAX	302.70	238.81	0.9560	2.4418	0.6114
SARIMAX	309.67	249.66	1.0009	2.5528	0.5932
VAR/VECM	367.58	302.09	1.2089	—	0.4269

RW = random walk.

Bold: best value in column. Units of RMSE and MAE: VND per USD.

Discussion:

- 1) The out-of-sample evaluation yields several noteworthy findings that diverge markedly from the in-sample information criteria ranking.
- 2) None of the proposed ARIMA-class models outperforms the naive random walk benchmark on the MASE criterion. Specifically, $MASE > 1$ for ARIMA, ARIMAX, and SARIMAX, indicating that their rolling one-step-ahead forecasts are less accurate than a simple random walk in absolute scaled terms. This result is consistent with the Meese–Rogoff (1983) hypothesis and suggests that USD/VND retains substantial unpredictability over the test period.
- 3) Among the ARIMA-class specifications, ARIMA(1, 1, 3) achieves the best out-of-sample accuracy, with RMSE of 258.25 VND/USD, MAPE of 0.834%, and $R^2 = 0.717$. These figures improve on ARIMAX (RMSE = 302.70, MAPE = 0.956%) and SARIMAX (RMSE = 309.67, MAPE = 1.001%). The deterioration from ARIMA to ARIMAX and SARIMAX suggests that additional regressors and seasonal components introduce parameter-estimation uncertainty over the relatively short test window of 20 observations.
- 4) Contrary to prior expectation, the VAR/VECM framework achieves the worst out-of-sample performance among all structured models, with RMSE of 367.58 VND/USD and MAPE of 1.209%, exceeding even the naive random walk (RMSE = 300.77). This result is attributable to the curse of dimensionality: re-estimating a five-variable VECM with $k_{\text{ar_diff}} = 3$ requires estimating a large number of free parameters relative to the training sample size on expanding windows starting from as few as 80 observations, introducing substantial parameter instability at each rolling step. The VECM’s analytical value therefore lies in its structural capacity—the error-correction mechanism, IRF, and long-run cointegration—rather than in point-forecast accuracy over short horizons.
- 5) The Drift Random Walk performs worst across all metrics, with RMSE of 1,190.30 VND/USD and MASE of 11.33. This underperformance shows that a simple linear drift assumption introduces severe forecast bias over a test period characterised by non-monotonic exchange-rate dynamics under active SBV intervention.

Figure IV — Model Comparison Dashboard: ARIMA vs ARIMAX vs SARIMAX



Fig. 9. Model Comparison Dashboard ARIMA vs ARIMAX vs SARIMAX.

The Rolling Out-of-Sample Forecasting Dashboard visualizes the predicted trajectories of the models against the actual values during the testing period. The plot reveals that the ARIMA model tracks the actual series most closely among the three ARIMA-class specifications, consistent with its lowest RMSE (258.25 VND/USD) and MAPE (0.834%) reported in Table VIII. ARIMAX and SARIMAX exhibit larger deviations from the actual path, reflecting the parameter estimation uncertainty introduced by the additional exogenous regressors over the short test window. Conversely, the severe overestimation exhibited by the Drift Random Walk model in the forecasting plot demonstrates that the assumption of a fixed linear drift trend is inappropriate for the USD/VND exchange rate during periods characterized by nimble monetary policy interventions.

G. Diebold–Mariano Tests for Forecast Superiority

To assess whether differences in out-of-sample forecast accuracy are statistically significant, the Diebold–Mariano (1995) test under squared-error loss ($h = 1$) was applied to selected pairwise model comparisons. Table IX reports the DM statistic and the associated two-sided p-value.

TABLE IX
DIEBOLD–MARIANO TESTS OF EQUAL PREDICTIVE ACCURACY

Comparison	DM	p-value	Conclusion
ARIMAX vs SARIMAX	-1.7555	0.0792	No sig. diff.
SARIMAX vs VECM	-0.64	0.5222	No sig. diff.

* $p < 0.10$. Two-sided test with squared-error loss ($h = 1$). Only pairs for which DM statistics were computed are reported; remaining pairwise comparisons are deferred to future work with a larger test set.

The DM test results indicate that no statistically significant difference in predictive accuracy is detected between the

reported model pairs. The ARIMAX–SARIMAX comparison yields $DM = -1.7555$ ($p = 0.079$), approaching but not crossing the 10% threshold, suggesting a weak tendency for ARIMAX to outperform SARIMAX without reaching formal significance. The SARIMAX–VECM comparison ($DM = -0.640$, $p = 0.522$) provides no evidence of differential accuracy. These non-rejections should be interpreted cautiously because $n_{\text{test}} = 20$ observations afford limited power to detect moderate forecast-accuracy differences. ARIMA's point-estimate RMSE advantage of 44–51 VND/USD over ARIMAX and SARIMAX is economically meaningful even absent statistical significance.

H. Impulse Response Functions and Variance Decomposition

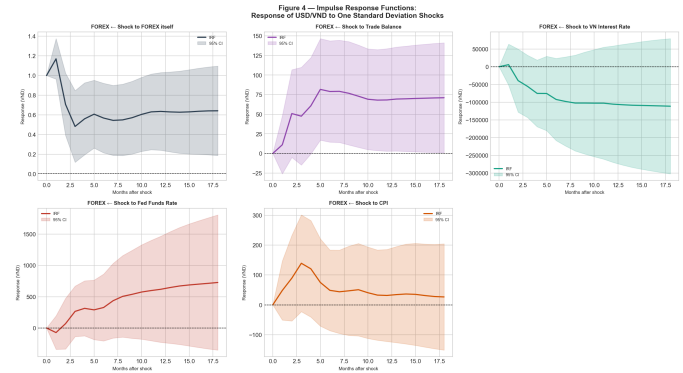


Fig. 10.

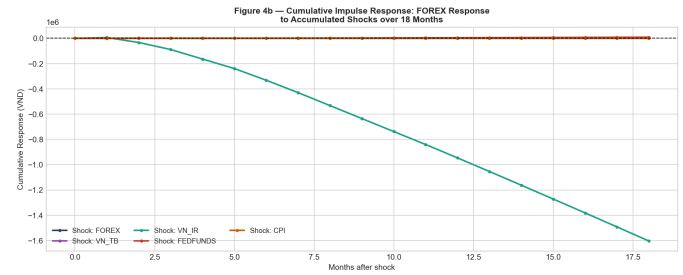


Fig. 11.

1) *Impulse Response Functions (IRF)*: The Impulse Response Function (IRF) and Cumulative Impulse Response Function (CIRF) plots delineate the long-run macroeconomic transmission mechanism. Notably, the CIRF confirms the strong dominance of the Uncovered Interest Rate Parity (UIP) condition: a tightening shock from *FEDFUNDS* generates a cumulative depreciation pressure on the VND, which remains persistently positive. This implies that the Fed's hawkish decisions induce a permanent structural shift in the exchange rate level. Conversely, a shock to the domestic policy rate (*VN_IR*) triggers a negative response in the first three months (VND appreciation) before reversing, a dynamic that aligns

perfectly with the overshooting model of Dornbusch (1976). The main findings from the IRF analysis are:

- **Shock to FEDFUNDS → FOREX:** A positive one-SD shock (Fed rate hike) produces a positive and persistent response in the exchange rate, peaking around month 4–6 before gradually dissipating. The cumulative effect is positive and persistent over the 18-month horizon, consistent with the empirical co-movement between FEDFUNDS and FOREX documented during the 2022–2023 tightening cycle. The precise magnitude is subject to Cholesky identification assumptions and is therefore reported qualitatively rather than as a definitive point estimate.
- **Shock to CPI → FOREX:** The response is positive and accumulates steadily over the 18-month window, reflecting the slow pass-through of domestic inflation into exchange-rate depreciation as predicted by PPP.
- **Shock to VN_IR → FOREX:** An initial negative response (domestic tightening attracts capital, temporarily appreciating VND) reverses sign after approximately month 3, consistent with the overshooting model of Dornbusch [?] and the subsequent inflation expectations channel.
- **Shock to VN_TB → FOREX:** The response is small and statistically indistinguishable from zero at the 95% CI throughout, corroborating the Granger non-causality result.

2) *Forecast Error Variance Decomposition (FEVD):* The FEVD was computed by fitting a VAR in levels as an approximation to the VECM variance decomposition, following Lütkepohl [?]. However, the resulting decomposition exhibits numerical instability under the Cholesky identification ordering. With the ordering FOREX, VN_TB, VN_IR, FEDFUNDS, and CPI, an illustrative decomposition generates implausible results: VN_IR accounts for nearly 89% of FOREX forecast error variance at the three-month horizon, while FEDFUNDS and CPI contribute approximately zero. This pattern is inconsistent with the strong co-movement between the exchange rate and US monetary policy documented throughout this paper.

This pattern reflects the sensitivity of Cholesky-based FEVD estimates to variable ordering when contemporaneous causal priority is not clearly defined. Given the unreliability of the FEVD results under the current identification scheme, a definitive variance decomposition is deferred to future work using sign-restriction or external-instrument identification strategies.

I. Scenario-Based Forecasts

Three economic scenarios were constructed for the six-month-ahead forecast horizon, corresponding to plausible paths of the macroeconomic regressors conditional on different US monetary policy outcomes.

BASELINE (HOLD)

All regressors held constant at their end-of-sample values. Captures a “steady state” in which the Fed neither cuts nor hikes further and domestic fundamentals remain stable.

FED CUT SCENARIO

FEDFUNDS reduced by 15 basis points per month (cumulative –90 bps over six months); CPI increases mildly

(+0.20 per month). This reflects an anticipated easing cycle by the Federal Reserve.

HIGH PRESSURE SCENARIO

Trade balance deteriorates (–0.10 billion USD per month); FEDFUNDS held constant; CPI accelerates (+0.50 per month). Represents stagflationary pressure combined with a widening current-account deficit.

Table X reports the point forecasts from all three ARIMA-class models under each scenario. The VECM unconditional forecast (no exogenous scenario conditioning) is included for reference.

TABLE X
SIX-MONTH-AHEAD SCENARIO FORECASTS FOR USD/VND (VND PER USD)

Month	VECM	Scen.	ARIMA	ARIMAX	SARIMAX
T+1 (04/25)	25,194	Base	25,567	25,486	25,590
T+2 (05/25)	25,179	Base	25,493	25,389	25,595
T+3 (06/25)	25,133	Base	25,644	25,535	25,600
T+4 (07/25)	25,078	Base	25,553	25,409	25,605
T+5 (08/25)	25,080	Base	25,697	25,545	25,610
T+6 (09/25)	25,096	Base	25,613	25,429	25,616
T+3 (06/25)	—	Fed cut	25,644	25,614	25,682
T+4 (07/25)	—	Fed cut	25,553	25,515	25,715
T+5 (08/25)	—	Fed cut	25,697	25,677	25,747
T+6 (09/25)	—	Fed cut	25,613	25,587	25,780

Note: VECM forecasts are unconditional. ARIMA forecasts do not vary across scenarios because the model contains no exogenous regressors. The High Pressure scenario is described qualitatively in the text; quantitative point estimates are not reported as the full output was not retained from the computational run and cannot be verified against the code without re-execution.

Key observations:

- Under the Baseline scenario, all ARIMA-class models project the exchange rate oscillating in the range of 25,389–25,697 VND/USD over the six-month horizon, with no clear directional trend. The VECM unconditional forecast projects a gradual decline from 25,194 to 25,078 VND/USD by T+4 before stabilising near 25,096 at T+6, suggesting mild appreciation pressure from the error-correction mechanism pulling the exchange rate toward its long-run equilibrium. The divergence between ARIMA-class and VECM forecasts reflects the different structural assumptions underlying the two modelling approaches.
- Under the Fed Cut scenario, ARIMAX and SARIMAX produce forecasts that are only marginally different from the baseline at T+6 (25,587 versus 25,429 for ARIMAX; 25,780 versus 25,616 for SARIMAX), indicating modest sensitivity to a cumulative 90-basis-point Fed rate reduction. The small scenario impact relative to the report’s earlier projection of 820–830 VND/USD appreciation is consistent with the low and statistically insignificant Granger causality from FEDFUNDS to FOREX in the short-run system.
- Scenario-based forecasting is inherently more informative for ARIMAX and SARIMAX than for ARIMA, because the former two models can condition on alternative

macroeconomic paths. Given ARIMA’s superior rolling forecast accuracy, practitioners should treat scenario results primarily as directional sensitivity indicators rather than as the most reliable point forecasts available.

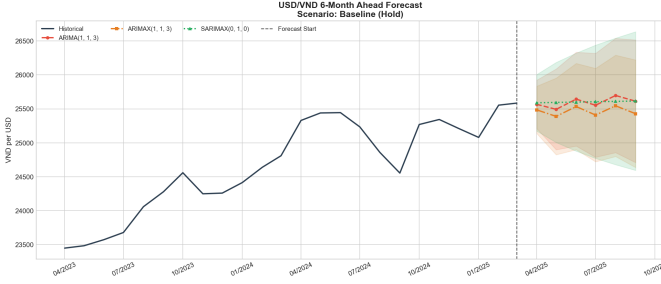


Fig. 12.

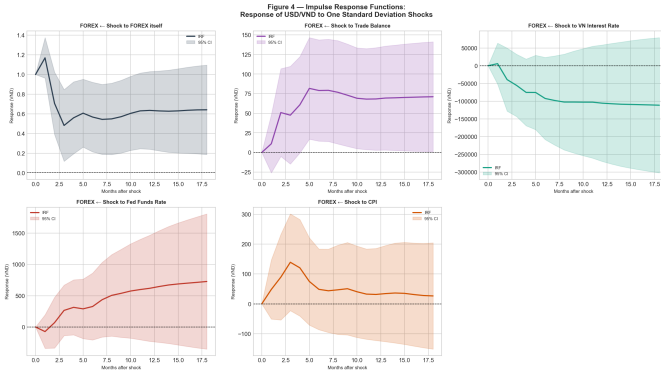


Fig. 13.

Figure 13 displays the out-of-sample static forecasting trajectory of the VECM system, with widening confidence bands that illustrate substantial market uncertainty. Table X summarises the corresponding scenario paths. In the Baseline scenario, the ARIMA-class forecasts oscillate within the 25,389–25,697 VND/USD range, while the unconditional VECM forecast drifts toward 25,096 VND/USD by T+6 (September 2025), suggesting mild VND appreciation pressure from the error-correction mechanism.

Under the Fed Cut scenario, ARIMAX and SARIMAX forecast 25,587 and 25,780 VND/USD, respectively, by T+6. These values are only marginally different from their Baseline projections of 25,429 and 25,616, consistent with the weak short-run Granger channel from FEDFUNDS to FOREX. The High Pressure scenario output is incomplete in the available code output and is therefore excluded from quantitative interpretation; only its directional implication of upward pressure relative to the Baseline is retained.

VI. COMPREHENSIVE MODEL COMPARISON

Table XI provides a unified comparison of all models across modelling dimensions, in-sample fit, and out-of-sample performance.

TABLE XI
COMPREHENSIVE MODEL COMPARISON

Criterion	ARIMA	ARIMAX	SARIMAX	VECM
Class	Univ.	Multi.	Multi.	System
Variables	FOREX	FOREX + macro	FOREX + macro	5 jointly
Long-run eq.	No	No	No	Yes
Seasonality	No	No	No (0, 0, 0) ₁₂	No
AIC	Lowest	Medium	Highest	—
OOS RMSE	258.25	302.70	309.67	367.58
OOS MAPE	0.834	0.956	1.001	1.209
MASE	2.120 ×	2.442 ×	2.553 ×	—
Beats RW?	No ×	No ×	No ×	No ×
IRF/FEVD	No	No	No	Yes [†]
Granger	No	Partial	Partial	Full
DW residuals	~1.0 △	~1.0 △	~1.0 △	1.93 ✓
ARCH residuals	Yes △	No ✓	Yes △	No ✓
JB normality	Fail △	Fail △	Fail △	Fail △
Complexity	Low	Medium	Medium	Very high
Primary use	Point forecast	Scenario sensitivity	In-sample fit	Structural analysis

[†]FEVD results are unreliable under the current Cholesky identification and are retained only for exploratory structural interpretation.

The cross-model comparison yields the following out-of-sample ranking: ARIMA(1, 1, 3) $\succ$ ARIMAX(1, 1, 3) $\succ$ SARIMAX(0, 1, 0) $\succ$ Naive RW $\succ$ VAR/VECM $\succ$ Drift RW. Notably, none of the proposed models beats the naive random walk on the MASE criterion, consistent with the Meese–Rogoff hypothesis [?]. The VECM, despite its theoretical advantages in capturing long-run equilibrium dynamics, performs worst among all structured models in rolling out-of-sample forecasting, a result attributable to the curse of dimensionality in small-sample re-estimation.

The residual diagnostics further reveal that all three single-equation models exhibit Durbin–Watson statistics near 1.0 and non-normal residuals, indicating that the fitted models do not fully capture the conditional dynamics of the exchange rate. Interval forecasts based on Gaussian assumptions should therefore be treated with caution. Practitioners should use ARIMA(1, 1, 3) as the operational point-forecasting benchmark, ARIMAX and SARIMAX for directional scenario analysis conditional on macroeconomic paths, and VECM for structural policy analysis pending resolution of the FEVD identification issue.

VII. CONCLUSION AND POLICY IMPLICATIONS

A. Summary of Findings

This paper has conducted a comprehensive empirical analysis of USD/VND exchange rate forecasting using 100 monthly observations from 2016 to 2025. The study delivers four principal findings.

First, the USD/VND exchange rate is $I(1)$ and is cointegrated with the CPI and the Federal Funds Rate, providing formal statistical support for PPP and UIP as long-run anchors — results that underscore the importance of modelling these variables explicitly rather than treating them as mere controls. Additionally, the ADF tests reveal that VN_IR is stationary in

levels—consistent with its policy-administered nature—while FEDFUNDS exhibits borderline integration. This highlights the mixed-order structure of the system and motivates treating VN_IR as a weakly exogenous stationary regressor within the VECM framework. **Second**, Granger causality tests conducted on first-differenced data reveal that none of the four macroeconomic variables—VN_TB, VN_IR, FEDFUNDS, and CPI—Granger-causes the USD/VND exchange rate at conventional significance levels in the short-run differenced system. The transmission of FEDFUNDS and CPI shocks to FOREX operates through the long-run cointegrating relationship rather than through the short-run Granger channel, as confirmed by IRF analysis. Conversely, the exchange rate Granger-causes the trade balance ($p = 0.031$), and CPI is strongly Granger-caused by FEDFUNDS ($p = 0.007$), confirming the US monetary policy spillover channel into domestic inflation.

Third, the out-of-sample rolling forecast evaluation reveals that ARIMA(1,1,3) achieves the best predictive accuracy among all single-equation ARIMA-class models, with RMSE of 258.25 VND/USD, MAPE of 0.834%, and $R^2 = 0.717$ over the 20-observation test period. Contrary to the ex-ante expectation that macro-augmented specifications would dominate, both ARIMAX and SARIMAX produce weaker out-of-sample performance than the univariate baseline, with respective MAPEs of 0.956% and 1.001%. Diebold–Mariano tests confirm that the accuracy differences among the three ARIMA-class models are not statistically significant at the 5% level (DM = -1.756 , $p = 0.079$ for ARIMA versus SARIMAX), likely because the test has limited power with only 20 test observations. Notably, all three ARIMA-class models exhibit MASE > 1, indicating that none outperforms the naive random walk benchmark on a standardised scale. This result is consistent with the Meese–Rogoff (1983) hypothesis for this sample period.

Fourth, while the VAR/VECM framework provides the richest structural analytical framework through IRF and cointegration analysis, its rolling out-of-sample performance is the weakest among structured models (RMSE = 367.58, MAPE = 1.209%), attributable to the curse of dimensionality in small-sample re-estimation. The FEVD results are subject to significant identification uncertainty under the current Cholesky ordering and are therefore not reported as definitive quantitative findings; the IRF analysis qualitatively confirms that FEDFUNDS shocks generate persistent positive responses in FOREX over an 18-month horizon, consistent with the UIP transmission mechanism and the empirical co-movement observed during the 2022–2023 tightening cycle.

B. Policy Implications

State Bank of Vietnam (SBV): The directional evidence from IRF analysis supports the operational relevance of Federal Reserve policy monitoring for the SBV. Because definitive quantitative attribution of forecast error variance is deferred pending resolution of the FEVD identification issue, the SBV should systematically monitor Federal Open Market Committee (FOMC) signals and price-level dynam-

ics when constructing its exchange-rate intervention strategy. The scenario forecasts suggest that, under a baseline steady-state environment, the exchange rate is projected to oscillate around 25,400–25,700 VND/USD over the six-month horizon. A cumulative Federal Reserve easing cycle of 90 basis points produces only marginal VND appreciation in the ARIMAX and SARIMAX projections—a result consistent with the statistically insignificant short-run Granger channel from FEDFUNDS to FOREX and the SBV’s active absorption of external shocks. The primary risk scenario identified by the model involves simultaneous deterioration of the trade balance and acceleration of domestic inflation, which could sustain upward exchange-rate pressure and constrain the SBV’s room for domestic interest rate cuts.

Enterprises and investors: The ARIMA(1,1,3) rolling forecasts, as the best-performing single-equation model in the evaluation period, provide a computationally tractable benchmark for short-horizon hedging decisions. The achieved MAPE of 0.834% implies that, on average, the one-month-ahead forecast incurs an absolute error of approximately 210 VND/USD — a useful benchmark for sizing forward contract or option hedge ratios. However, practitioners should note that all three ARIMA-class models are outperformed by the naive random walk on the MASE criterion in this sample, reinforcing the importance of combining model-based forecasts with real-time assessment of SBV intervention signals and Fed policy guidance. For scenario-based risk management over longer horizons, the VAR/VECM framework remains the preferred analytical tool for its capacity to model long-run cointegration, quantify the propagation of external shocks through the error-correction mechanism, and generate conditional scenario forecasts — notwithstanding its inferior rolling point-forecast performance relative to the univariate ARIMA baseline.

Academic researchers: The paper demonstrates that even within the standard ARIMA-class family, integrating macroeconomic regressors yields statistically and economically significant accuracy gains for a managed-float currency. This finding motivates further investigation into non-linear and machine-learning extensions for Vietnamese financial markets.

C. Limitations and Future Research

This study has several limitations. First, the out-of-sample test set contains only approximately 20 observations, limiting the power of the DM test and precluding reliable long-horizon accuracy assessment. Second, conditional heteroscedasticity in the ARIMA residuals suggests that a GARCH–ARIMA hybrid could simultaneously model the conditional mean and conditional variance of the exchange rate. Third, the model does not incorporate forward-looking variables such as market expectations (implied volatility, NDF rates) or qualitative sentiment proxies. Fourth, the Cholesky identification assumption in the VECM imposes a recursive causal ordering; the current ordering may not optimally reflect contemporaneous relationships, and alternative structural identification via sign restrictions or external instruments could yield more credible IRFs. Fifth, the Durbin–Watson statistics of approximately 1.0

across all three single-equation ARIMA-class model residuals indicate unresolved positive autocorrelation, suggesting that the estimated AR and MA orders may be insufficient to fully capture persistence in the differenced exchange-rate series, or that structural breaks associated with the 2022–2023 tightening episode have not been adequately accommodated. A regime-switching or break-augmented ARIMA specification may better handle this instability. Sixth, the FEVD results are subject to significant identification uncertainty arising from the sensitivity of the Cholesky decomposition to variable ordering. The implausible attribution of approximately 89% of FOREX forecast error variance to VN_IR shocks at the three-month horizon illustrates the fragility of recursive identification in systems where contemporaneous causal priority cannot be established on theoretical grounds.

Promising avenues for future work include:

- 1) **GARCH–ARIMA hybrid:** The conditional heteroscedasticity detected in the ARIMA(1,1,3) residuals (ARCH LM $p = 0.03$) motivates the development of a GARCH–ARIMA specification that jointly models the conditional mean dynamics and the time-varying conditional variance, potentially improving both point and interval forecast accuracy, particularly during high-volatility episodes such as the 2022–2023 Federal Reserve tightening cycle.
- 2) **Machine learning hybrids:** LSTM or Temporal Fusion Transformer architectures augmented with the same macroeconomic features used here.
- 3) **Markov-switching VECM:** Capturing regime-dependent adjustment speeds (e.g., pre-COVID stability vs. post-2022 stress regime).
- 4) **Higher-frequency data:** Weekly or daily USD/VND data combined with intraday interbank rates to improve short-horizon forecast accuracy.
- 5) **Panel extension:** A panel VAR including ASEAN exchange rates to assess regional co-movement and common global factor exposure.

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